Form Approved
OMB No. 0704-0188

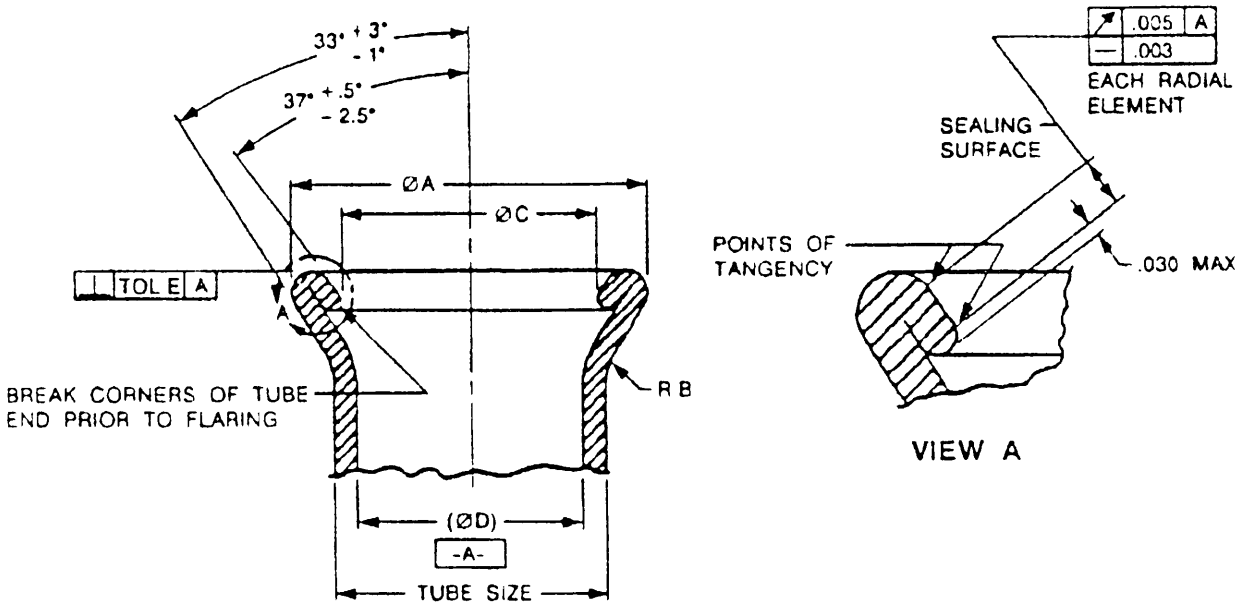


FIGURE 1. FLARED TUBE END

TABLE I. DIMENSIONS, TUBE END

DASH NUMBER	TUBE SIZE NOMINAL OD	A DIA +.000 -.010	B RAD ±.010	C DIA MAX	C DIA MIN	D DIA (REF)	NOMINAL WALL THICKNESS	E
2	.125	.200	.032	.096 .086	.062 .048	.062 .048	.028 .035	.002
3	.188	.302		.168 .158	.124 .110	.124 .110	.028 .035	.003
4	.250	.359		.230 .219	.187 .173	.187 .173	.028 .035	
5	.313	.421		.292 .281 .259	.250 .235 .205	.250 .235 .205	.028 .035 .049	.004
6	.375	.484	.046	.354 .344 .322	.312 .298 .268	.312 .298 .268	.028 .035 .049	

1. FLARE SEALING SURFACE SHALL BE SMOOTH, UNIFORM CONICAL SURFACE FREE FROM PITS, CRACKS, NICKS, FLAT SPOTS, BURRS, DENTS, SCRATCHES, CHATTER OR LONGITUDINAL AND SPIRAL TOOL MARKS THAT MIGHT RESULT IN A LEAK PATH. DISCONTINUITIES SHALL NOT EXCEED:

   a. 10% OF THE NOMINAL TUBE WALL THICKNESS FOR TUBES SPECIFYING PROOF PRESSURE TEST UNDER 200 PSI.

   b. 5% OF THE NOMINAL TUBE WALL THICKNESS FOR TUBES SPECIFYING PROOF PRESSURE TEST AT 200 PSI OR GREATER.

2. NON-SEALING SURFACE OF FLARE SHALL BE FREE OF CRACKS AND PIT MARKS.

© ENTIRE STANDARD REVISED

"DESIGN STANDARD"

INCH-POUND

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY-AT NAVY-AS AIR FORCE-99 REVIEW: AF-82, ARMY-AV, DLA-CS USER: ARMY-ME PROJECT NUMBER: 4730-0054	MILITARY SPECIFICATION SHEET TITLE: TUBING END DOUBLE FLARE, STANDARD DIMENSIONS FOR	SPECIFICATION SHEET NUMBER: MS33583 REV C DATE: 28 JUNE 1993 SUPERSEDING: MS33583B 25 AUG 1988 AMSC: N/A FSC: 4730	

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3. DEFINITIONS:

CRACK	–	A PARTING OR CLEFT DUE TO THE FRACTURE OF SOLID METAL DURING METAL FORMING.
DENTS & CREASES	–	A HOLLOW OR A DEPRESSION IN THE TUBE WALL. THE SURFACE OF THE WALL MATERIAL IS NOT CUT OR REMOVED.
BURRS, GOUGES & NICKS	–	A DEFECT WHEREIN THE SURFACE OF THE MATERIAL ITSELF IS ACTUALLY BROKEN OR REMOVED SUFFICIENTLY TO CAUSE LOCAL THINNING, USUALLY THE BURR WILL HAVE SHARP EDGES AND CORNERS.
SCRATCH	–	A MARK WHICH MAY RUN IN ANY DIRECTION AND IS, OR APPEARS TO BE, A CUT OR TEAR IN THE SURFACE MATERIAL.
WRINKLE & FOLDS	–	A RADIAL OR LONGITUDINAL DEFORMATION. RADIAL WRINKLES ARE MEASURED FROM PEAK TO VALLEY. LONGITUDINAL WRINKLES ARE MEASURED ACCORDING TO THE HEIGHT ABOVE THE NATURAL HEIGHT OF THE SURFACE.
TOOL & HANDLING MARKS	–	LONGITUDINAL OR CIRCUMFERENCIAL GROOVES OF SHALLOW DEPTH PRODUCED BY MANUFACTURING TOOLS GRIPPING OR MOVING OVER THE SURFACE OF THE TUBE DURING FORMING.
FLAT	–	A SHALLOW POCKET OR HOLLOW ON THE SURFACE OF THE SEALING SURFACE OR TUBE, DUE TO NONFILLING OF METAL DURING FORMING.

4. FLARE SEALING SURFACE SHALL BE 32 MICROINCHES R_a PER ANSI/ASME B46.1.

5. INTENDED TO BE USED WITH ALUMINUM TUBING AND STANDARD FLARED TUBE FITTINGS

6. **UNLESS OTHERWISE SPECIFIED**, DIMENSIONS ARE IN INCHES

7. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M

8. EXAMPLE OF CALLOUT: "END FLARED PER MS33583-6, .028 WALL"

MS33583 –6 .028 WALL

- MS33583 — BASIC NUMBER
- –6 — DASH NUMBER; NOMINAL TUBE SIZE IN .0625 INCREMENTS (.375 DIAMETER)
- .028 WALL — TUBE NOMINAL WALL (SEE TABLE I)

(C) ENTIRE STANDARD REVISED

PREPARING ACTIVITY: NAVY – AS CUSTODIANS: ARMY – AT NAVY – AS AIR FORCE – 99 REVIEW: AF – 82, ARMY – AV, DLA – CS USER: ARMY – ME PROJECT NUMBER: 4730 – 0254	MILITARY SPECIFICATION SHEET TITLE: TUBING END DOUBLE FLARE, STANDARD DIMENSIONS FOR	SPECIFICATION SHEET NUMBER: MS33583 REV C DATE: 28 JUNE 199 SUPERSEDING: MS33583B 25 AUG 1988 AMSC: N/A FSC: 4730	
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